

about 10,000 times higher than regular commercial cameras. Dynamic range is a measure of the brightest and darkest areas a camera can capture simultaneously. With these, the camera can see better in driving conditions such as the transition from a dark tunnel into bright sunlight, or during hazy or foggy conditions.

According to *Optica*, the camera would enable cars to detect hazards, other cars and people three times farther away than colour cameras used in cars today.

In addition to automotive applications, researchers are exploring using the camera to detect cancerous cells, which exhibit a different light polarisation than normal tissue, and to improve ocean exploration.

Graphene could exceed bandwidth demands of future telecommunications

Researchers from Graphene Flagship project—one of the biggest research initiatives of European Commission—have showed that integrated graphene-based photonic devices offer a unique solution for next-generation optical communications. They have demonstrated how properties of graphene enable ultra-wide bandwidth communications coupled with low power consumption to radically change the way data is transmitted across optical communications systems.

This research could make graphene-integrated devices the key ingredient in the evolution of 5G, the Internet of Things (IoT) and Industry 4.0.

Recycled photovoltaic waste to boost circular economy

European Union-funded Horizon2020 CABRISS project has developed methods for recovering valuable materials from PV waste such as silicon (Si) for re-use in the industry. It has helped transform legal obligations under WEEE directive into new business opportunities by pioneering a circular economy based on recycled, reused and recovered indium (In), silicon and silver (Ag) materials for PV and other applications.

Project coordinator David Pelletier says, “CABRISS focuses mainly on the PV production value chain, demonstrating cross-sectorial industrial symbiosis with closed-loop processes.” Industrial symbiosis describes a network of diverse organisations for fostering eco-innovation, long-term culture change, and improving business and technical processes. CABRISS developed this process by providing raw materials as feed stocks for other industries.

Researchers used three different sources of PV waste in the project. The first involved a novel technique for delaminating and recovering all high-value materials like silver, silicon, indium, and high-purity glass from PV end-of-life thin film and silicon-based PV modules.

The second comprised solid waste from PV production, consisting of a mixture of broken silicon wafers and cells.

The final source is dry silicon powder PV production waste, known as kerf, recovered from materials lost during the cutting process.

They used laser technology to open the thin-film PV modules without damage, resulting in higher value for the recycled glass. “For silicon-based PV modules, an innovative and water-based technology was developed that does not break glass, resulting in the collection of all materials in silicon PV modules,” adds Pelletier.

Printable solar materials could turn parts of a house into solar panels

A team of researchers led by Derya Baran of King Abdullah University of Science and Technology’s (KAUST’s) Solar Center, Saudi Arabia, has developed a PV organic material that captures light efficiently and that potentially could be coated on building materials.



Derya Baran (left) and team have developed a PV organic material that captures light efficiently and could be coated on building materials

Baran explains that traditional rooftop solar panels are made from silicon, but organic molecules can also capture energy from sunlight. These molecules could be formulated as inexpensive printable inks that are applied to regular building components such as windows. Turning sunlight into electricity is a multi-step process, and the key to developing high-performance organic PV materials has been to find organic molecules that are good at every step.

When light strikes an organic PV material, it knocks free an electron, leaving behind a positively-charged hole. If the oppositely-charged electron and hole recombine, captured energy is lost. Thus, organic solar cells incorporate a mixture of electron donor and electron acceptor molecules to draw the charges apart.